



THE KING'S SCHOOL

2004
Higher School Certificate
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value



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2004 Higher School Certificate Trial Examination

Mathematics

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(c), (d), (f)		(b)	(a)		(e)	12
2			(b)		(a)		12
3		(c)	(b)			(a)	12
4			(c)	(a)			12
5					(a)	(b)	12
6			(c)	(a)		(b)	12
7			(a)		(c)	(b)	12
8	(a)				(b)		12
9	(a)	(b)					12
10	(i)			(ii), (iii), (v)	(iv)		12
Marks	20	10	26	20	26	18	120

Total marks – 120

Attempt Questions 1-10

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- | | |
|--|---|
| (a) Find the value of $\sin 2x$ if $x = 0.006$, correct to 2 significant figures. | 2 |
| (b) State the domain and range of the function $y = \log_e x$ | 2 |
| (c) Find $\lim_{x \rightarrow 0} \frac{x^3 - 3x}{6x}$ | 2 |
| (d) Solve the equation $(2x + 3)^2 = 4$ | 2 |
| (e) Find a primitive function of $(2x + 3)^4$ | 2 |
| (f) For what value of x do $y = 1 - 2x$ and $2y = 7 + 6x$ hold simultaneously? | 2 |

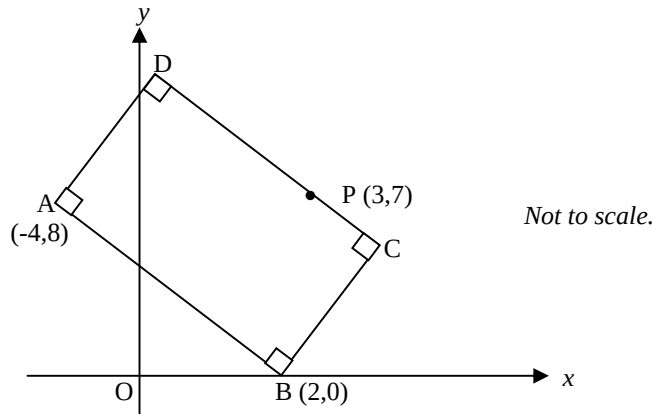
End of Question 1

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the gradient of the normal to the curve $y = \frac{x}{x^2 + 1}$ at the point $(0, 0)$ **3**

(b)



In the diagram, ABCD is a rectangle where $A = (-4, 8)$ and $B = (2, 0)$. $P(3, 7)$ is a point on the side CD.

- (i) Find the gradient of line AB. **1**
- (ii) Deduce that the equation of line AB is $4x + 3y - 8 = 0$ **2**
- (iii) Hence, or otherwise, show that the length of side BC is 5 units. **2**
- (iv) Write down the mid-point of side AB and deduce that $P(3, 7)$ is the mid-point of side CD. **2**
- (v) Write down the point of intersection of the diagonals AC and BD. **1**
- (vi) Find the coordinates of point D. **1**

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.**Marks**

(a) Find the exact value of

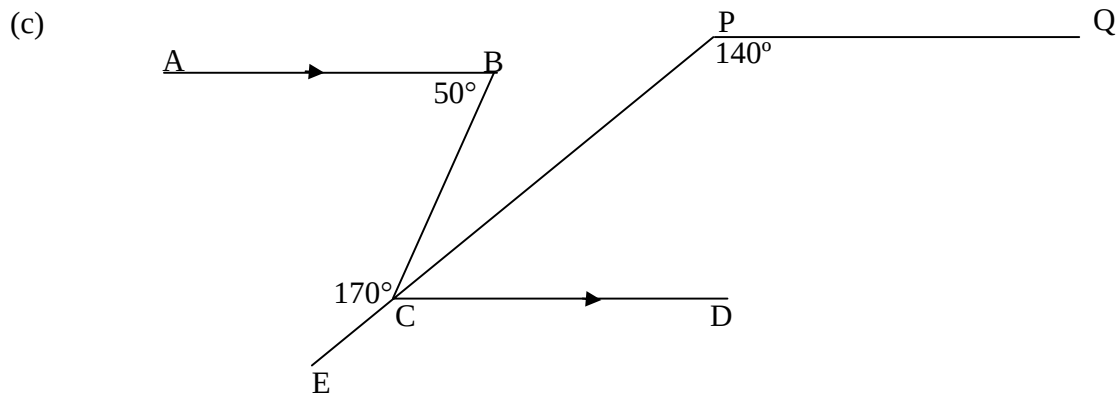
(i) $\int_0^{\frac{\pi}{6}} \sin x \, dx$ 2

(ii) $\int_0^1 \frac{6x^2}{x^3 + 1} \, dx$ 2

(b) For what values of k is $x^2 + 2x + k$

(i) concave upward? 1

(ii) positive for all values of x ? 2



In the diagram, $AB \parallel CD$ and ECP is a straight line.

$$\angle ABC = 50^\circ, \angle ECB = 170^\circ, \angle QPC = 140^\circ$$

(i) Find $\angle BCP$, giving reasons. 1

(ii) Find $\angle ECD$, giving reasons. 2

(iii) Prove that $PQ \parallel AB$ 2

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

-
- (a) (i) Sketch the curve $y = \tan \frac{x}{2}$ for $-2\pi \leq x \leq 2\pi$ **2**
- (ii) State the period of $y = \tan \frac{x}{2}$ **1**
- (iii) Solve the equation $\tan \frac{x}{2} = 1$ for $-2\pi \leq x \leq 2\pi$ **2**
- (b) Consider the two arithmetic series
- $$A = 11 + 13 + 15 + \dots$$
- and $B = -14 - 11 - 8 - \dots$
- (i) Find the sum of the first 40 terms of series A **1**
- (ii) The two series have the same number of terms and the same sum. How many terms are in the series? **3**
- (c) Find the values of a , b , c if
- $$a(x+1)^2 + b(x+1)(x-1) + c(x-1) \equiv x^2 + 7x$$
- 3**

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Consider the function $f(x) = x^4 - 6x^2 + 8x$

(i) Show that $f'(x) = 4(x+2)(x-1)^2$ 2

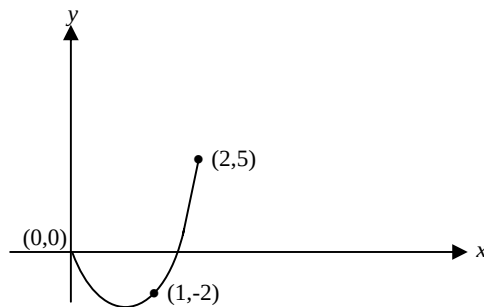
(ii) For what values of x is the function increasing? 2

(iii) Show that there is a minimum turning point at $x = -2$ 2

(iv) Show that there is a horizontal point of inflection at $x = 1$ 2

(b) (i) Briefly explain why Simpson's Rule gives the exact value of $\int_a^b f(x) dx$ if $f(x)$ is a quadratic function. 1

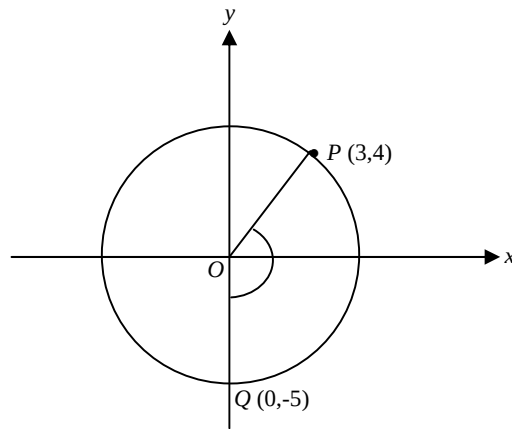
(ii)



A parabola $y = f(x)$ passes through the points $(0,0)$, $(1,-2)$ and $(2,5)$. Find the value of $\int_0^2 f(x) dx$ 3

End of Question 5

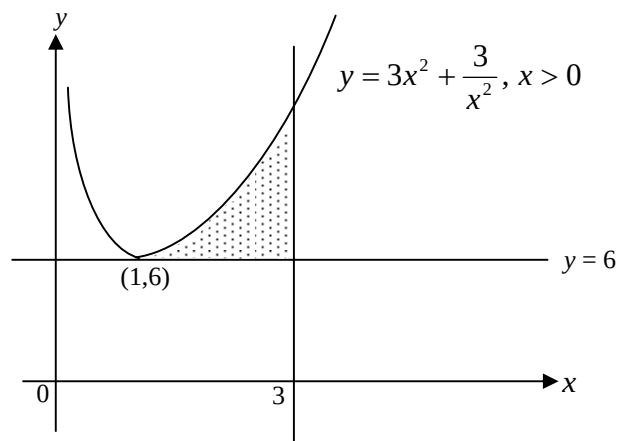
(a)



In the diagram, $P(3,4)$ and $Q(0,-5)$ are points on the circle $x^2 + y^2 = 25$.
Let $\angle POQ = \theta$, as marked on the diagram.

- | | |
|---|----------|
| (i) Find the length of chord PQ . | 1 |
| (ii) Show that $\cos \theta = -\frac{4}{5}$ | 2 |
| (iii) Find the area of minor sector POQ correct to 1 decimal place. | 2 |

(b)



In the diagram, the shaded region is bounded by the curve $y = 3x^2 + \frac{3}{x^2}$, $x > 0$,
and the two lines $y = 6$ and $x = 3$. Find the area of this shaded region.

4

Question 6 continues next page

Question 6 (continued)

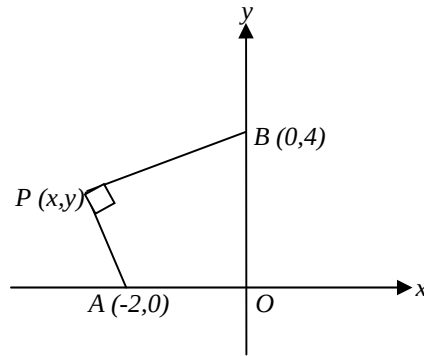
(c) The directrix of a parabola is the x axis and the focus is the point $(0,4)$.

(i) Write down the focal length of the parabola. **1**

(ii) Find the equation of the parabola. **2**

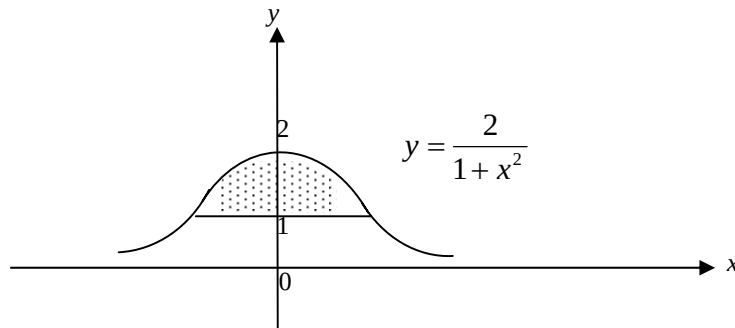
End of Question 6

- (a) $A(-2,0)$ and $B(0,4)$ are two points in the number plane. $P(x,y)$ is any point such that AP is perpendicular to BP .



- (i) Prove that the equation of the locus of $P(x,y)$ is $x(x+2) + y(y-4) = 0$ 3
- (ii) Deduce that the equation in (i) represents a circle and find its centre and radius. 3

(b)



Consider the region bounded by the curve $y = \frac{2}{1+x^2}$ and the line $y = 1$ as shown in the diagram.

The region is revolved about the y axis. Find the volume of the solid of revolution generated.

4

Question 7 continues next page

- (c) The population, P , of a rural town is growing exponentially according to the equation $P = 5000e^{0.1t}$, t measured in years. Currently, i.e. $t = 0$, the population is increasing at a rate of 500 people/year.

What rate of increase, correct to the nearest hundred, is expected after 20 more years?

2

End of Question 7

Question 8 (12 marks) Use a SEPARATE writing booklet.

Marks

-
- (a) Simplify the expression $x^{-1}y^2\left(x^{\frac{1}{2}} - y^{-1}\right)\left(x^{\frac{1}{2}} + y^{-1}\right)$, giving your answer with positive indices. 2
- (b) A particle moves on a straight line so that its velocity, $v \text{ m/s}$, at any time t seconds is given by $v = (t-1)^4 + \frac{t}{2}$, $t \geq 0$
- (i) Find the initial velocity and show that the particle never stops. 2
- (ii) Find the initial acceleration of the particle. 2
- (iii) Find the least value of the velocity. 2
- (iv) Sketch the velocity-time graph. 2
- (v) Find the distance travelled by the particle in the first 2 seconds. 2

End of Question 8

Question 9 (12 marks) Use a SEPARATE writing booklet.

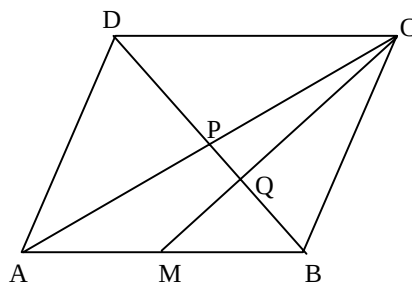
Marks

- (a) In the land of Welf the interest on investments is paid continuously. If an interest rate of $100k\%$ p.a. is given then it can be shown that the amount in the fund after t years, $t \geq 0$, is $A(t)$ where

$$A(t) = Pe^{kt}, \quad P \text{ is the initial investment.}$$

- | | | |
|-------|---|----------|
| (i) | A particular fund, F , in Welf pays 10% p.a. interest. Show that $k = 0.1$ | 1 |
| (ii) | Sally invests \$5000 into fund F for 20 years. Find, correct to the nearest dollar, the amount Sally would have after the 20 years. | 1 |
| (iii) | Sally wants to be more wealthy in Welf and decides not to terminate her investment of \$5000 until it has grown to at least \$100 000. For how many years will she need to wait? | 2 |
| (iv) | Lucy also invests in fund F . She decides to deposit \$1500 into the fund at the start of each year for 20 years. Find, correct to the nearest dollar, the amount Lucy would have after the 20 years. | 3 |

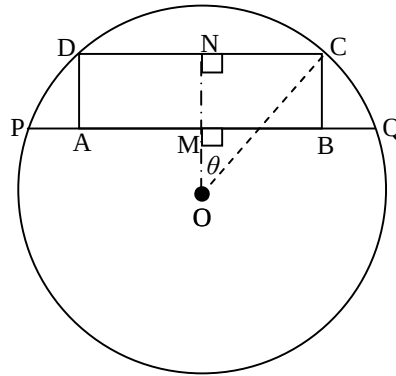
(b)



In the diagram, $ABCD$ is a parallelogram. The diagonals AC and BD meet at P . M is the mid-point of AB and MC meets BD at Q .

- | | | |
|-------|---|----------|
| (i) | Show that $\triangle CDQ$ is similar to $\triangle MBQ$ | 1 |
| (ii) | Deduce that $DQ = 2BQ$ | 2 |
| (iii) | Prove that $BQ = 2QP$ | 2 |

End of Question 9



In the diagram, O is the centre of a circle of radius $\sqrt{6}$ cm and PQ is a chord of length 4 cm. $ABCD$ is a rectangle constructed in the minor segment cut off by chord PQ . OM is drawn perpendicular to PQ so that M is the mid-point of both chord PQ and side AB . N is the mid-point of side CD .

Let $\angle CON = \theta$, θ in radians.

- | | |
|---|----------|
| (i) Show that $OM = \sqrt{2}$ cm | 1 |
| (ii) Show that $0 < \theta < 1$ | 2 |
| (iii) Show that the area, a , of rectangle $ABCD$ is given by
$a = 4\sqrt{3} \sin \theta (\sqrt{3} \cos \theta - 1)$ | 3 |
| (iv) Show that $\frac{da}{d\theta} = 4\sqrt{3}(2\sqrt{3} \cos^2 \theta - \cos \theta - \sqrt{3})$ | 3 |
| (v) Find the maximum area of the rectangle. | 3 |

End of Examination

TKS MATHEMATICS TRIAL 2004 SOLUTIONS

Qn 1 (a) $\sin 0.012 = 0.012$, 2 sig. figs

(b) Domain $x > 0$, Range all real values for y

(c) $\lim_{x \rightarrow 0} \frac{x(x^2-3)}{6x} = \lim_{x \rightarrow 0} \frac{x^2-3}{6} = -\frac{1}{2}$

or $\lim_{x \rightarrow 0} \left(\frac{x^2}{6} - \frac{3}{6} \right) = -\frac{1}{2}$

(d) $2x+3=2$ or $2x+3=-2$

$\therefore x = -\frac{1}{2}$ or $-\frac{5}{2}$

(e) $\frac{(2x+3)^5}{5 \times 2} = \frac{(2x+3)^5}{10}$

(f) $2(1-2x) = 7+6x$

$2-4x = 7+6x$

$\therefore 10x = -5$

$x = -\frac{1}{2}$

Qn 2

$$(a) \frac{dy}{dx} = \frac{(x^2+1)1 - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$$

$$\text{At } (0,0), \frac{dy}{dx} = 1$$

$\therefore$ gradient of normal at $(0,0)$ is -1

$$(b) (i) \text{ gradient } AB = \frac{0-8}{2-4} = -\frac{8}{-2} = 4$$

$$(ii) \text{ line } AB \text{ is } y = -\frac{4}{3}(x-2)$$

$$\text{or } 3y = -4x + 8$$

$$\text{ie } 4x + 3y - 8 = 0$$

(iii) BC = perpendicular distance from P to line AB

$$\therefore BC = \frac{|12 + 21 - 8|}{\sqrt{4^2 + 3^2}} = \frac{25}{5} = 5$$

$$(iv) M_{AB} = \left(-\frac{2}{2}, \frac{8}{2}\right) = (-1, 4)$$

$\therefore$ From (iii), if $PM_{AB} = 5$ then P is mid-point of CD

$$PM_{AB} = \sqrt{4^2 + 3^2} = 5 \quad \therefore \text{result.}$$

(v) Diagonals meet at mid-point of PM_{AB} , from (iv)

$$\text{ie at } \left(\frac{2}{2}, \frac{11}{2}\right) = (1, \frac{11}{2}) = Q, \text{ say}$$

(vi) [LOTS OF WAYS]

$$\text{e.g. } B \rightarrow Q = Q \rightarrow D$$

$$\Rightarrow D = (0, 11)$$

Qn 3

$$(a) (i) \left[-\cos x \right]_0^{\frac{\pi}{6}} = -\frac{\sqrt{3}}{2} - -1 = 1 - \frac{\sqrt{3}}{2}$$

$$(ii) = 2 \int_0^1 \frac{3x^2}{x^3+1} dx = 2 \left[\ln(x^3+1) \right]_0^1 \\ = 2 \ln 2$$

(b) (i) $\cup$ $\therefore$ all values of k

$$(ii) \text{ Need } \Delta < 0 \Rightarrow 4 - 4k < 0 \\ \text{or } 4k > 4 \text{ or } k > 1$$

(c) (i) $\angle ECP = 180^\circ$, ECP a straight line

$$\therefore \angle BCP = 10^\circ$$

(ii) $\angle BCD = \angle ABC = 50^\circ$, alternate $\angle$ s in $\parallel$ lines AB, CD

$$\therefore \angle ECD = 360^\circ - (170^\circ + 50^\circ), \text{ angles at point } C \\ = 140^\circ$$

(iii) From (i & ii), $\angle QPC = \angle ECD = 140^\circ$ and these
- are corresponding angles

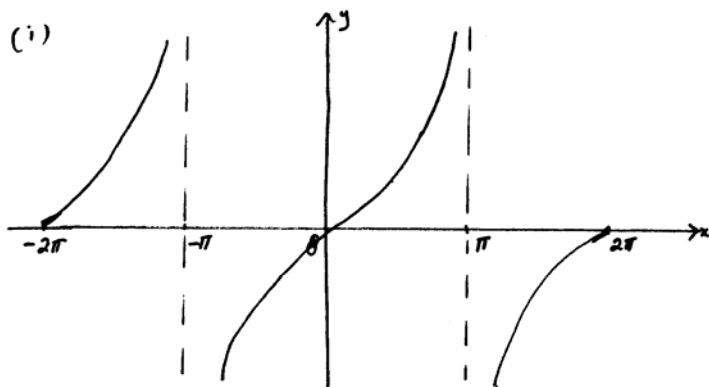
$$\therefore PQ \parallel CD$$

But $AB \parallel CD$

$$\therefore PQ \parallel AB$$

Qn 4

(a) (i)



(ii) period = 2π

(iii) $\therefore \frac{x}{2} = \frac{\pi}{4} \text{ or } -\frac{3\pi}{4} \text{ for } -\pi \leq \frac{x}{2} \leq \pi$

[Sketch helps]

$$\Rightarrow x = \frac{\pi}{2} \text{ or } -\frac{3\pi}{2}$$

(b) (i) $S_{40} = \frac{40}{2} [22 + 39 \times 2] = 20 \times 100 = 2000$

(ii) $\therefore \frac{n}{2} (22 + (n-1)2) = \frac{n}{2} (-28 + (n-1)3)$

$$\Rightarrow 22 + 2n - 2 = -28 + 3n - 3$$

$$\therefore n = 51 \text{ i.e. there's 51 terms}$$

(c) Put $x=1$, $\therefore 4a = 1+7 \Rightarrow a=2$

$\therefore$ Equating coefficients of x^2 , $2+b=1$

$$\Rightarrow b=-1$$

Put $x=-1$, $\therefore -2c = 1-7$

$$\therefore c=3$$

[LOTS OF WAYS, of course]

Qn 5

(a) (i) $f'(x) = 4x^3 - 12x + 8 = 4(x^3 - 3x + 2)$

Now, $(x+2)(x-1)^2 = (x+2)(x^2 - 2x + 1)$
 $= x^3 - 2x^2 + x + 2x^2 - 4x + 2$
 $= x^3 - 3x + 2$

$\therefore f'(x) = 4(x+2)(x-1)^2$

(ii) A test for increasing function is $f'(x) > 0$

$4(x+2)(x-1)^2 > 0 \Rightarrow x+2 > 0$ since $4(x-1)^2 \geq 0$ for all x

$\therefore f(x)$ is increasing for $x > -2$

[Note If $x=1$, $f'(x)=0$ but the function is still increasing.
A curve is increasing if as x increases, $f(x)$ increases]

(iii) $f'(-2) = 0 \therefore$ at $x=-2$ there's a stationary point

But for $x < -2$, $f'(x) < 0$
& $x > -2$, $f'(x) > 0 \Rightarrow$ 

$\therefore$ at $x=-2$ there's a minimum turning point

(iv) $f'(x) = 4(x^3 - 3x + 2)$

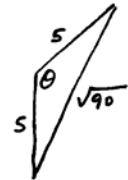
$\therefore f''(x) = 4(3x^2 - 3) = 12(x^2 - 1)$

$\therefore f''(1) = 0$ and $f''(0) < 0$, $f''(2) > 0$
change in concavity
and, as well, $f'(1) = 0$

$\therefore$ at $x=1$ there's a horiz. pt. of inflection

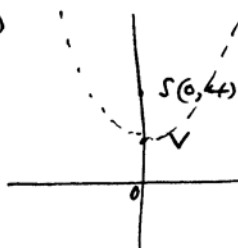
Qn 6

(a) (i) $PQ = \sqrt{3^2 + 9^2} = \sqrt{90}$

(ii)  $\therefore \cos \theta = \frac{25 + 25 - 90}{2 \times 5 \times 5} = \frac{-40}{50} = -\frac{4}{5}$

(iii) Area = $\frac{1}{2} \times 5^2 \times \theta$ where $\cos \theta = -\frac{4}{5}$
 $= \frac{1}{2} \times 25 \times 2.498 \dots = 31.2 \text{ u}^2, 1 \text{ d.p.}$

(b) $A = \int_1^3 3x^2 + \frac{3}{x^2} - 6 \, dx$
 $= \left[x^3 - \frac{3}{x} - 6x \right]_1^3$
 $= 27 - 1 - 18 - (1 - 3 - 6) = 16 \text{ u}^2$

(c)  (i) focal length $a = 2$
(ii) Vertex = $(0, 2)$
 $\therefore$ equation is $(x-0)^2 = 4 \times 2 (y-2)$
 $\therefore x^2 = 8(y-2)$

Qn 7

(a) (i) Since $AP \perp BP$ the product of their gradients is -1

$$\therefore \frac{y}{x+2} \times \frac{y-4}{x} = -1$$

$$\text{or } y(y-4) = -x(x+2)$$

$$\text{or } x(x+2) + y(y-4) = 0$$

$$(ii) \therefore x^2 + 2x + y^2 - 4y = 0$$

$$\Rightarrow (x+1)^2 - 1 + (y-2)^2 - 4 = 0$$

$$\text{or } (x+1)^2 + (y-2)^2 = 5$$

is a circle, centre $(-1, 2)$, radius $\sqrt{5}$

$$(b) V = \pi \int_1^2 x^2 dy \text{ where } y = \frac{2}{1+x^2}$$

$$\therefore 1+x^2 = \frac{2}{y} \text{ or } x^2 = \frac{2}{y} - 1$$

$$\therefore V = \pi \int_1^2 \left(\frac{2}{y} - 1 \right) dy$$

$$= \pi \left[2 \ln y - y \right]_1^2$$

$$= \pi (2 \ln 2 - 2 - (0 - 1)) = \pi (2 \ln 2 - 1) \text{ m}^3$$

$$(c) \frac{dP}{dt} = 5000 \times 0.1 e^{0.1t} = 500 e^{0.1t}$$

$$\therefore \text{when } t = 20, \frac{dP}{dt} = 500 e^2 \approx 3700 \text{ people/yr,}$$

nearest hundred.

Qn 8

$$(a) \quad x^{-1} y^2 (x - y^{-2}) = y^2 - x^{-1} = y^2 - \frac{1}{x}$$

$$\left[\text{or } \frac{y^2 x - 1}{x} \right]$$

$$(b) \quad (i) \quad t=0, \quad v = (-1)^4 + 0 = 1 \text{ m/s}$$

$$\text{Now, } (t-1)^4 \geq 0 \text{ for all } t \geq 0$$

$$\text{and } \frac{t}{2} > 0 \text{ for all } t > 0$$

$$\therefore v > 0 \text{ for all } t \geq 0 \text{ i.e. particle never stops}$$

$$(ii) \quad \ddot{x} = \frac{dv}{dt} = 4(t-1)^3 + \frac{1}{2}$$

$$\therefore \text{ when } t=0, \quad \ddot{x} = -4 + \frac{1}{2} = -3\frac{1}{2} \text{ m/s}^2$$

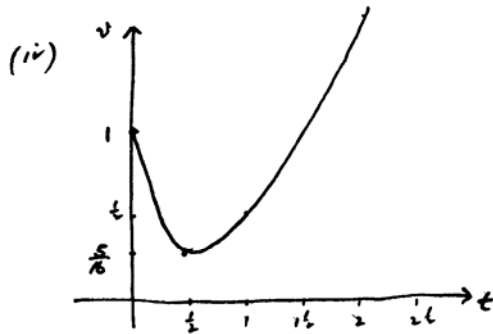
$$(iii) \quad \text{From (i) and (ii), least } v \text{ occurs when } \ddot{x} = 0$$

$$\Rightarrow 4(t-1)^3 + \frac{1}{2} = 0$$

$$\text{or } (t-1)^3 = -\frac{1}{8}$$

$$\therefore t-1 = -\frac{1}{2} \text{ or } t = \frac{1}{2}$$

$$\therefore \text{ least } v = \frac{1}{16} + \frac{1}{4} = \frac{5}{16} \text{ m/s}$$



$$(v) \quad x = \int_0^t (t-1)^4 + \frac{t}{2} dt = \left[\frac{(t-1)^5}{5} + \frac{t^2}{4} \right]_0^t$$

$$= \frac{1}{5} + 1 - \left(-\frac{1}{5} + 0 \right) = 1\frac{2}{5} \text{ m}$$

Qn 9

(a) (i) $\therefore 100k = 10 \Rightarrow k = 0.1$

(ii) $A(20) = 5000 e^{0.1 \times 20} = 5000 e^2$
 $= \$36945$, nearest dollar

(iii) Hae, $A(t) = 5000 e^{0.1t} \geq 100000$

$$\therefore e^{0.1t} \geq 20$$

$$\text{or } 0.1t \geq \ln 20$$

$$\therefore t \geq \frac{\ln 20}{0.1} = 29.957 \dots$$

$\Rightarrow$ Sally needs to wait approx 30 years

(iv) Lucy would have

$$1500 e^{0.1 \times 20} + 1500 \times e^{0.1 \times 19} + \dots + 1500 e^{0.1 \times 1}$$

$$= 1500 e^1 + 1500 e^2 + 1500 e^3 + \dots + 1500 e^2,$$

is a geometric series where $a = 1500 e^1$
 $r = e^1$

$$\therefore \text{Lucy would have } \frac{1500 e^1 ((e^1)^{20} - 1)}{e^1 - 1}$$

$$= \frac{1500 e^1 (e^2 - 1)}{e^1 - 1}$$

$$= \$100707, \text{ nearest dollar}$$

Q9 (b)

(i) In Δ s CDQ , MBQ

$$\angle CDQ = \angle MBQ, \text{ alternate } \angle \text{ s in } \parallel \text{ lines } CD, BA$$

$$\angle DCQ = \angle BMQ, \quad "$$

$\therefore \Delta CDQ \parallel \Delta MBQ$, 2 angles equal

(ii) From (i), $\frac{DQ}{BQ} = \frac{CD}{BM}$, ratios of corresponding sides

$$= 2, \text{ since } M \text{ is the mid-point of } AB, AB = CD$$

$$\therefore DQ = 2 BQ$$

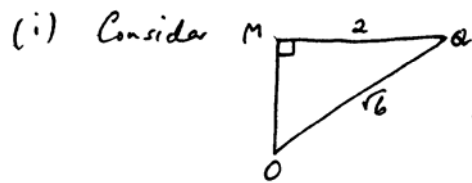
(iii) Now, $BQ + QP = PD$, diagonals of $\parallel$ gram bisect each other

$$= DQ - QP$$

$$= 2BQ - QP, \text{ (ii)}$$

$$\therefore BQ = 2QP$$

Qn 10

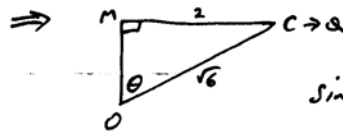


$$\therefore OM^2 + 4 = 6$$

$$\Rightarrow OM = \sqrt{2}$$

(ii) as C approaches the vertical position above N then $\theta \rightarrow 0$,
otherwise $\theta > 0$

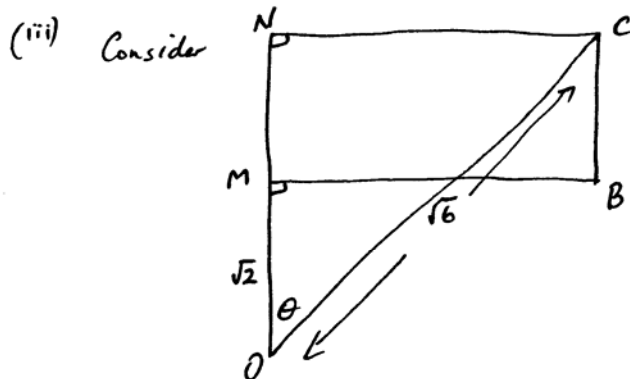
max θ occurs as C approaches Q



$$\sin \theta = \frac{2}{\sqrt{6}}$$

$$\Rightarrow \theta = 0.955 \dots < 1$$

$$\therefore 0 < \theta < 1$$



$$\text{Then } \sin \theta = \frac{NC}{\sqrt{6}}, \quad NC = \sqrt{6} \sin \theta$$

$$\therefore DC = 2\sqrt{6} \sin \theta$$

$$\text{and } \cos \theta = \frac{ON}{\sqrt{6}} \quad \therefore ON = \sqrt{6} \cos \theta$$

$$\therefore MN = \sqrt{6} \cos \theta - \sqrt{2} = CB$$

$$\begin{aligned} \therefore \text{Area, } a &= 2\sqrt{6} \sin \theta (\sqrt{6} \cos \theta - \sqrt{2}) \\ &= 2\sqrt{6} \sqrt{2} \sin \theta (\sqrt{3} \cos \theta - 1) \\ &= 4\sqrt{3} \sin \theta (\sqrt{3} \cos \theta - 1) \end{aligned}$$

$$(iv) \frac{da}{d\theta} = 4\sqrt{3} \left(\sin\theta (-\sqrt{3} \sin\theta) + (\sqrt{3} \cos\theta - 1) \cos\theta \right)$$

$$= 4\sqrt{3} \left(-\sqrt{3} \sin^2\theta + \sqrt{3} \cos^2\theta - \cos\theta \right)$$

$$= 4\sqrt{3} \left(-\sqrt{3} (1 - \cos^2\theta) + \sqrt{3} \cos^2\theta - \cos\theta \right)$$

$$= 4\sqrt{3} \left(2\sqrt{3} \cos^2\theta - \cos\theta - \sqrt{3} \right)$$

$$(v) \frac{da}{d\theta} = 4\sqrt{3} \left((\sqrt{3} \cos\theta + 1) (2\cos\theta - \sqrt{3}) \right)$$

$$= 0 \text{ only if } 2\cos\theta - \sqrt{3} = 0 \text{ since } 0 < \theta < 1$$

and since as $\theta \rightarrow 0$ or $\theta \rightarrow 1$, then $a \rightarrow 0$,

the value of θ from $2\cos\theta - \sqrt{3} = 0$ must produce
a maximum turning point $\Rightarrow$ maximum area

$$\therefore \text{max area occurs when } \cos\theta = \frac{\sqrt{3}}{2} \text{ and } \therefore \sin\theta = \frac{1}{2}$$

$$\therefore \text{max } a = 4\sqrt{3} \cdot \frac{1}{2} \left(\sqrt{3} \cdot \frac{\sqrt{3}}{2} - 1 \right) \text{ cm}^2$$

$$= \sqrt{3} \text{ cm}^2$$